

Student Name: Alex Tonev

Student ID: 201895897

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Report on Sandwich Ontology Design

The designed ontology attempts to model the components and structure of a sandwich as described by the [Wikipedia article](#), as per the project guidelines.

At first the article describes a sandwich to be a piece of food consisting of one or more fillings between two pieces of bread. It mentions a wide variety of possible fillings, as well as the possibility for there to be a number of pieces of bread different than two, introducing the concepts of a “sandwich deck”, and that of an “open sandwich”. Multi-deck sandwiches may introduce multiple additional layers of bread and filling, while open sandwiches omit the need for a top bread. These special cases make the initial proposition for the definition of a sandwich being “bread-filling-bread” obsolete, as the number of pieces of bread is no longer fixed.

To accommodate all special cases, the ontology was designed to think of sandwiches in a different way. The core of an open sandwich – our base case – is a base of bread, and some fillings. This open sandwich can then become closed with a further addition of a top bread, or be complemented by a second deck. An interesting observation can be made that even sandwiches with the original 2 slices of bread can have some of their fillings on top, as are the cases with the [Croque monsieur](#) and the Portuguese [Francesinha](#). The ontology views those cases as “open two-deck sandwiches”, treating the de-facto closing bread with additional toppings simply as a second deck.

The ontology includes the following ten classes:

- **Sandwich**: the root class.
- **Deck**: composing of Bread and Filling.
- **Bread**, with subclass **ClosingBread**.
- **Filling**, with five subclasses: **Meat**, **Cheese**, **Veg**, **Spread**, and **OtherProtein**.

The “ClosingBread” subclass was chosen for specificity in object properties we will later see, but also because many times the top bread of a sandwich has particularities, as is the case with [hamburgers](#) for example.

The Filling classes chosen were all top-level fillings, with the intent of covering as much ground as possible. “Spread” can be applied to both sweet and savory spreads.

“OtherProtein” is an attempt to fit in miscellaneous items in sandwiches serving that particular purpose, such as eggs, tofu, or hashbrowns. No hierarchy was applied there, because of the constraint of there being up to five object properties.

All five object properties were needed to describe a sandwich, given the above classes:

1. hasDeck (Sandwich → Deck): must have at least one deck to be a sandwich.
2. hasBaseBread (Deck → Bread): each Deck must have a base slice.
3. hasFilling (Deck → Filling): a deck must contain one or more fillings.

4. hasClosingBread (Sandwich $\rightarrow$ ClosingBread): optional top bread.
5. isOnTopOf (Deck $\rightarrow$ Deck): to model the vertical order of decks in a sandwich.

This provides a comprehensive list of the possible relationship between all classes, as they were defined. Sandwiches are comprised of at least one deck and an optional closing bread. Any deck has a base and a filling. The order of decks is modelled with the “isOnTopOf” relationship.

To complete the ontology, the above structural restrictions are formalized as a set of axioms:

1. Any sandwich must contain at least one deck:
- `Sandwich  $\sqsubseteq$  ( $\geq 1$  hasDeck.Deck)`
2. A sandwich may contain at most one closing bread:
- `Sandwich  $\sqsubseteq$  ( $\leq 1$  hasClosingBread.ClosingBread)`
3. Every deck needs to have exactly one base bread:
- `Deck  $\sqsubseteq$  (=1 hasBaseBread.Bread)`
4. Every deck must contain one or more fillings:
- `Deck  $\sqsubseteq$  ( $\geq 1$  hasFilling.Filling)`
5. Two decks cannot be on top of one another at the same time:
- `Asymmetric(isOnTopOf)`
6. No deck can be on top of itself:
- `Irreflexive(isOnTopOf)`

With this, the ontology provides a comprehensive structure for describing most sandwiches, within the required number of classes and properties. The design allows for the inference of whether a sandwich is open or closed, as well as knowledge of how the decks are stacked, without allowing for an obvious paradox to occur. The modular design of the ontology allows for the addition of further classes, such as a open and closed sandwiches, and the addition of hierarchy within the Filling top class.

With the constraint on the number of properties allowed, however, some obvious limitations arised. The “Filling” set of subclasses don’t have a relationship to encode the order they appear in, as is possible with decks. Additionally, there are no clear restrictions on two individuals using the same component instance, like sharing a closing bread for example.

References

Use of AI:

1. Use of ChatGPT 4o, as a thinking partner during ontology design.

“should sandwich hasClosingBread point to bread or closingbread?”

“hmm if it is a reverse I continue thinking this isn't the right move. Help me with possible properties I might not have thought of so far”

2. Use of ChatGPT 4o, for help with structuring during writing

“Knowing what you know now comprise a full breakdown on what to write in the report”

“So far I have this. How do I transition to axioms from here. The designed ontology attempts to model the components and structure of a sandwich as described by the Wikipedia article, as per the project guidelines. At first the article describes a sandwich to be a piece of food consisting of one or more...”